Genome-wide fusion-hotspot analysis of msk_impact_50k_2026

Abstract

This manuscript synthesizes a genome-wide fusion-hotspot cohort scan of msk_impact_50k_2026: 3919 genes carried at least one structural-variant record in the cohort, of which 544 passed the >= 5-distinct-patient recurrence gate. All 544 gated genes were attempted: 4 using hand-curated gene configs and 520 auto-configured, with 20 gated-in gene(s) left unresolvable; 523 of the 544 attempted genes were successfully analyzed with the full registered algorithm suite. 4 genes reached genome-wide Benjamini-Hochberg FDR significance (q < 0.05): NTRK3 (58 events, 94.8% in-frame, 94.8% domain-retained, Fisher p=0.0695489, q=8.1653e-09); ROS1 (122 events, 59.0% in-frame, 87.7% domain-retained, Fisher p=0.214047, q=0.000221281); ETV6 (90 events, 71.1% in-frame, 75.6% domain-retained, Fisher p=5.12326e-06, q=0.00233279); FLI1 (118 events, 61.9% in-frame, 5.9% domain-retained, Fisher p=0.988837, q=0.00820622). 15 additional genes form a highly ranked non-FDR-significant tier flagged for targeted follow-up (see Honorable mentions, below).

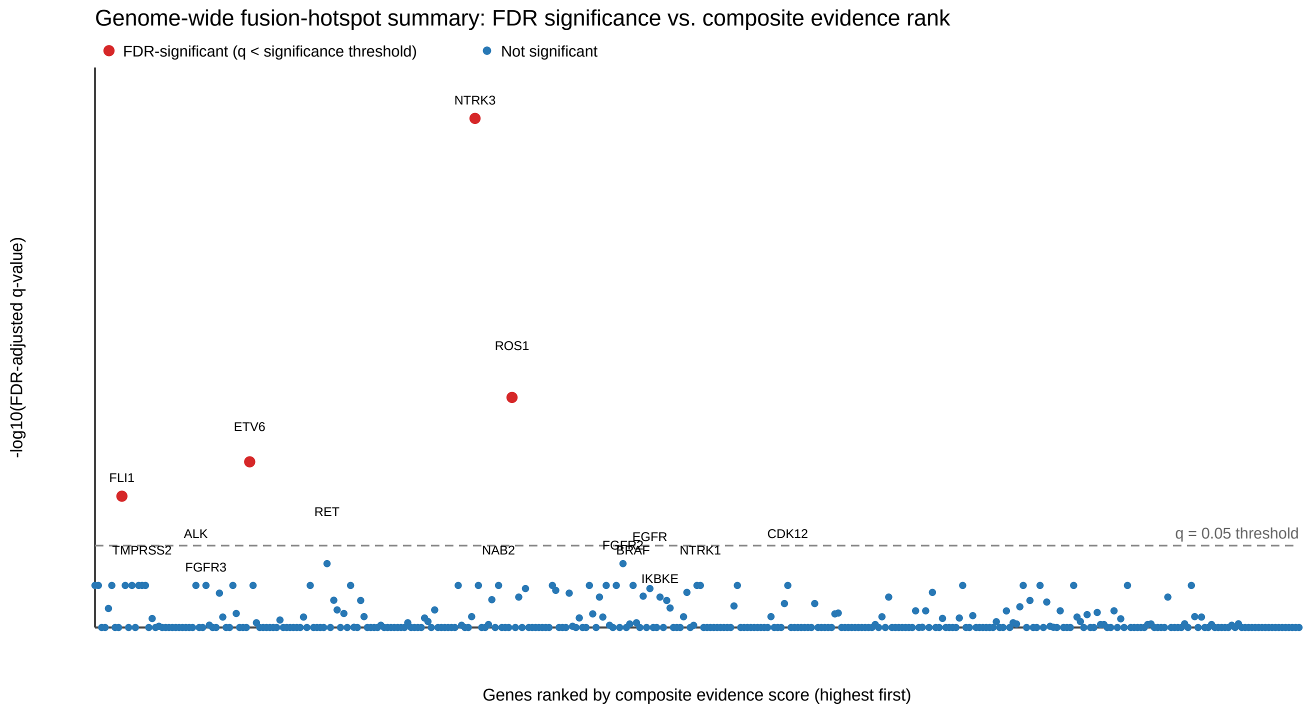
Methods

Structural-variant records were retrieved from the msk_impact_50k_2026 cBioPortal study and gated to genes with at least 5 distinct patient(s) carrying a structural-variant record (544 of 3919 genes passed the gate). All 544 gated genes were attempted: 4 using hand-curated gene configs and 520 auto-configured from Genome Nexus canonical-transcript/Pfam-domain data, with 20 gated-in gene(s) left unresolvable; 523 of the 544 attempted genes were successfully analyzed. Each successfully analyzed gene was run through the algorithm suite recorded for this scan (composite_score, confidence_stats, cutpoint_detection, domain_disruption, domain_retention, exon_retention, frequency, joint_partner). Domain-retention and domain-disruption significance were assessed per gene with Fisher's exact test and a breakpoint-position permutation test; the resulting p-values across the 359 genes that produced at least one computable p-value were jointly corrected with Benjamini-Hochberg false-discovery-rate correction at q < 0.05.

Results

Genome-wide summary

Genome-wide summary of 359 scanned genes with an FDR-adjusted q-value, ranked left-to-right by composite evidence score; the dashed line marks the q=0.05 significance threshold, with 4 genes above it.



Genome-wide fusion-hotspot summary: FDR significance vs. composite evidence rank
FDR-significant (q < significance threshold)
Not significant
NTRK3
ROS1
ETV6
FLI1
RET
ALK
TMPRSS2
FGFR3
NAB2
FGFR2
EGFR
BRAF
NTRK1
IKBKE
CDK12
q = 0.05 threshold
-log10(FDR-adjusted q-value)
Genes ranked by composite evidence score (highest first)

FDR-significant and honorable-mention genes

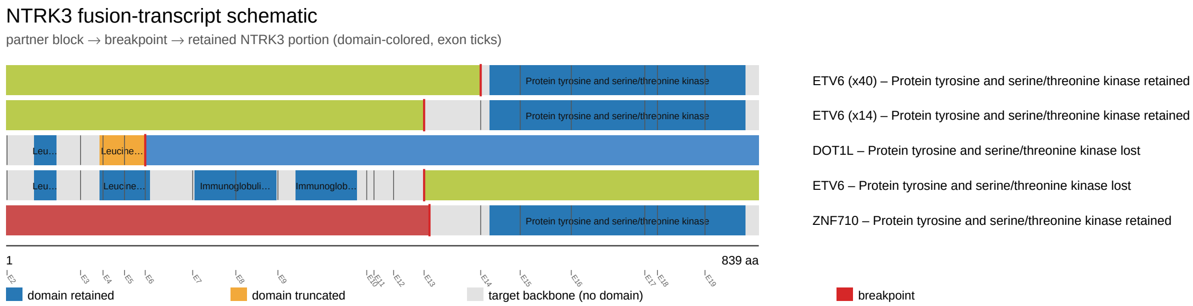
Gene	Tier	N events	In-frame%	Domain-ret.%	Fisher p	Best q
NTRK3	FDR-significant	58	94.83	94.83	0.06955	8.165e-09
ROS1	FDR-significant	122	59.02	87.7	0.214	0.0002213
ETV6	FDR-significant	90	71.11	75.56	5.123e-06	0.002333
FLI1	FDR-significant	118	61.86	5.932	0.9888	0.008206
RET	Honorable mention	194	75.26	92.27	0.0004197	0.09669
FGFR2	Honorable mention	136	79.41	84.56	0.0004247	0.09669
ALK	Honorable mention	272	83.09	95.96	0.001917	0.2147
EGFR	Honorable mention	55	50.91	74.55	0.01145	0.2407
BRAF	Honorable mention	179	84.36	91.06	0.01337	0.2147
FGFR3	Honorable mention	152	48.68	93.42	0.01948	0.2147
CDKN2B	Honorable mention	12	16.67	16.67	0.02222	0.3614
IKBKE	Honorable mention	12	25	33.33	0.02424	0.3288
TMPRSS2	Honorable mention	867	29.99	6.113	0.0292	0.2147
NTRK1	Honorable mention	78	41.03	82.05	0.04826	0.2147
INPPL1	Honorable mention	14	28.57	50	0.04895	0.4687
CDK12	Honorable mention	43	16.28	37.21	0.0677	0.2147
NAB2	Honorable mention	72	34.72	68.06	0.1126	0.2147
KDM5A	Honorable mention	10	30	50	0.119	0.8787
FH	Honorable mention	16	6.25	12.5	0.1333	0.6719

Gene highlights

NTRK3 (FDR-significant)

NTRK3 was analyzed across 58 fusion events, 94.8% in-frame and 94.8% domain-retained. Domain-retention Fisher's exact test p=0.0695489 (raw not statistically significant at alpha=0.05). Genome-wide BH-adjusted q=8.1653e-09 (reaches genome-wide FDR significance).

Reused from the NTRK3 individual gene report (fusion_schematic.svg).

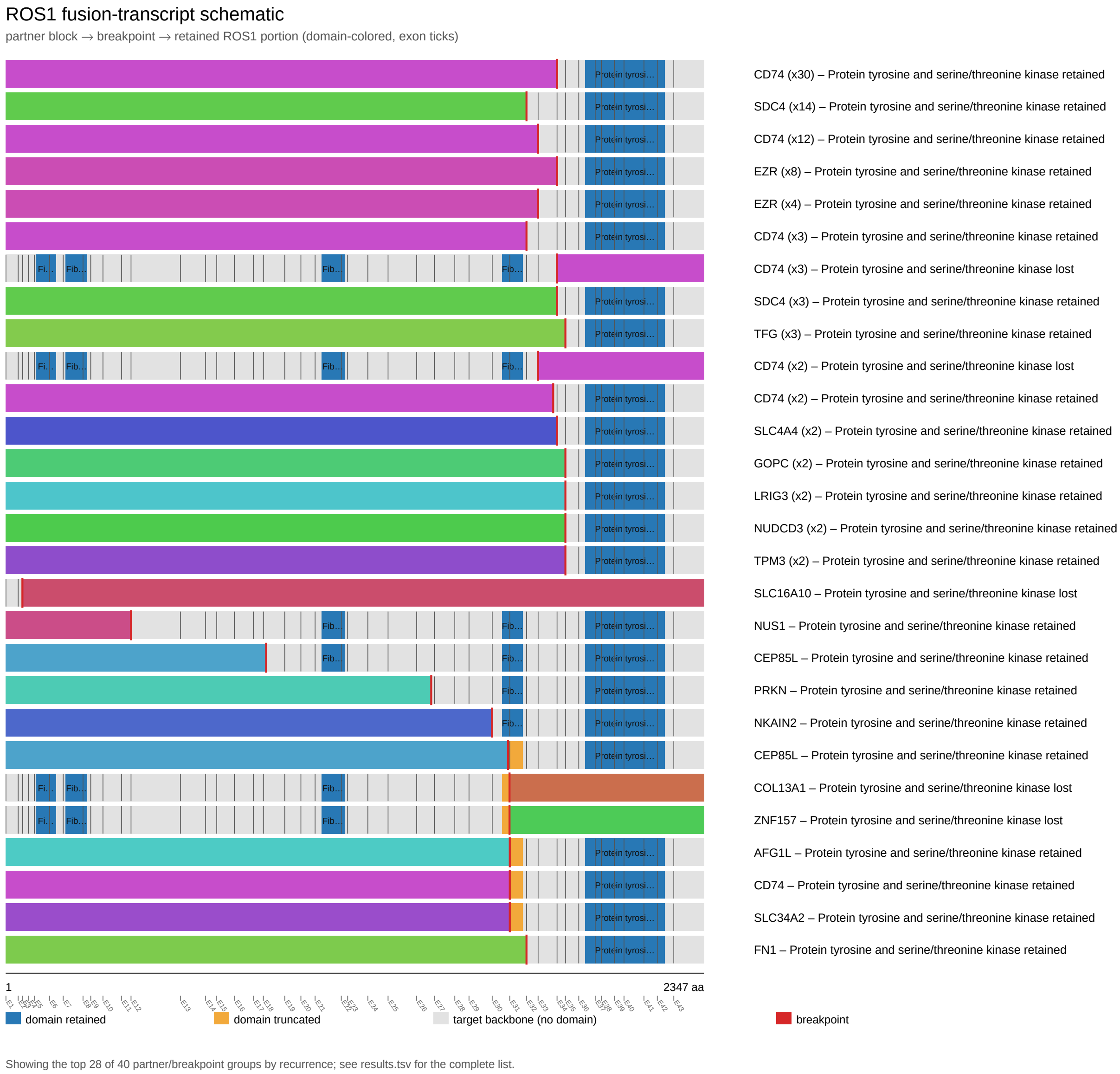


Full per-gene detail: gene_reports/ntrk3/report.md

ROS1 (FDR-significant)

ROS1 was analyzed across 122 fusion events, 59.0% in-frame and 87.7% domain-retained. Domain-retention Fisher's exact test p=0.214047 (raw not statistically significant at alpha=0.05). Genome-wide BH-adjusted q=0.000221281 (reaches genome-wide FDR significance).

Reused from the ROS1 individual gene report (fusion_schematic.svg).

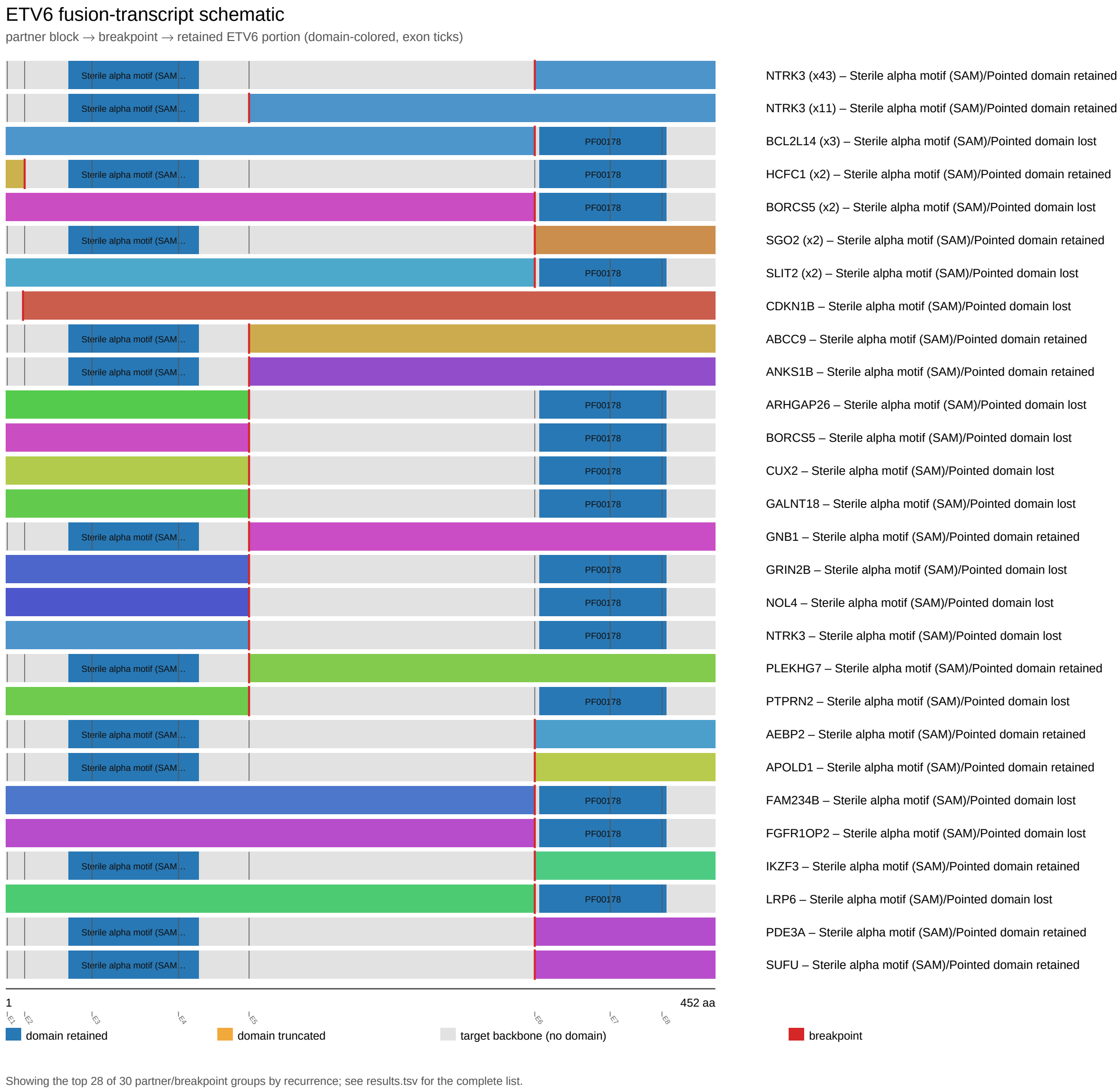


Full per-gene detail: gene_reports/ros1/report.md

ETV6 (FDR-significant)

ETV6 was analyzed across 90 fusion events, 71.1% in-frame and 75.6% domain-retained. Domain-retention Fisher's exact test p=5.12326e-06 (raw statistically significant at alpha=0.05). Genome-wide BH-adjusted q=0.00233279 (reaches genome-wide FDR significance).

Reused from the ETV6 individual gene report (fusion_schematic.svg).

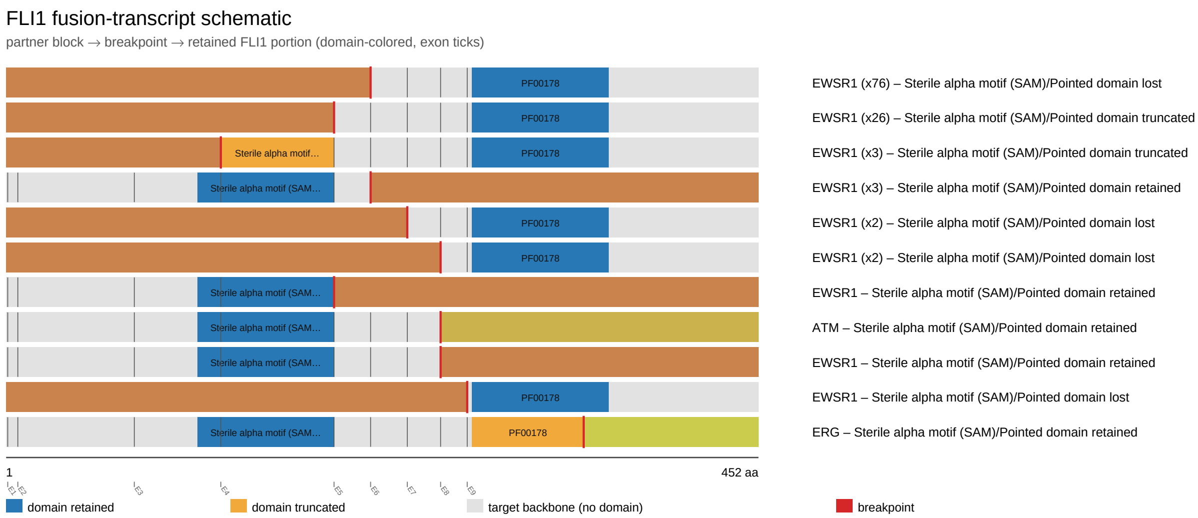


Full per-gene detail: gene_reports/etv6/report.md

FLI1 (FDR-significant)

FLI1 was analyzed across 118 fusion events, 61.9% in-frame and 5.9% domain-retained. Domain-retention Fisher's exact test p=0.988837 (raw not statistically significant at alpha=0.05). Genome-wide BH-adjusted q=0.00820622 (reaches genome-wide FDR significance).

Reused from the FLI1 individual gene report (fusion_schematic.svg).

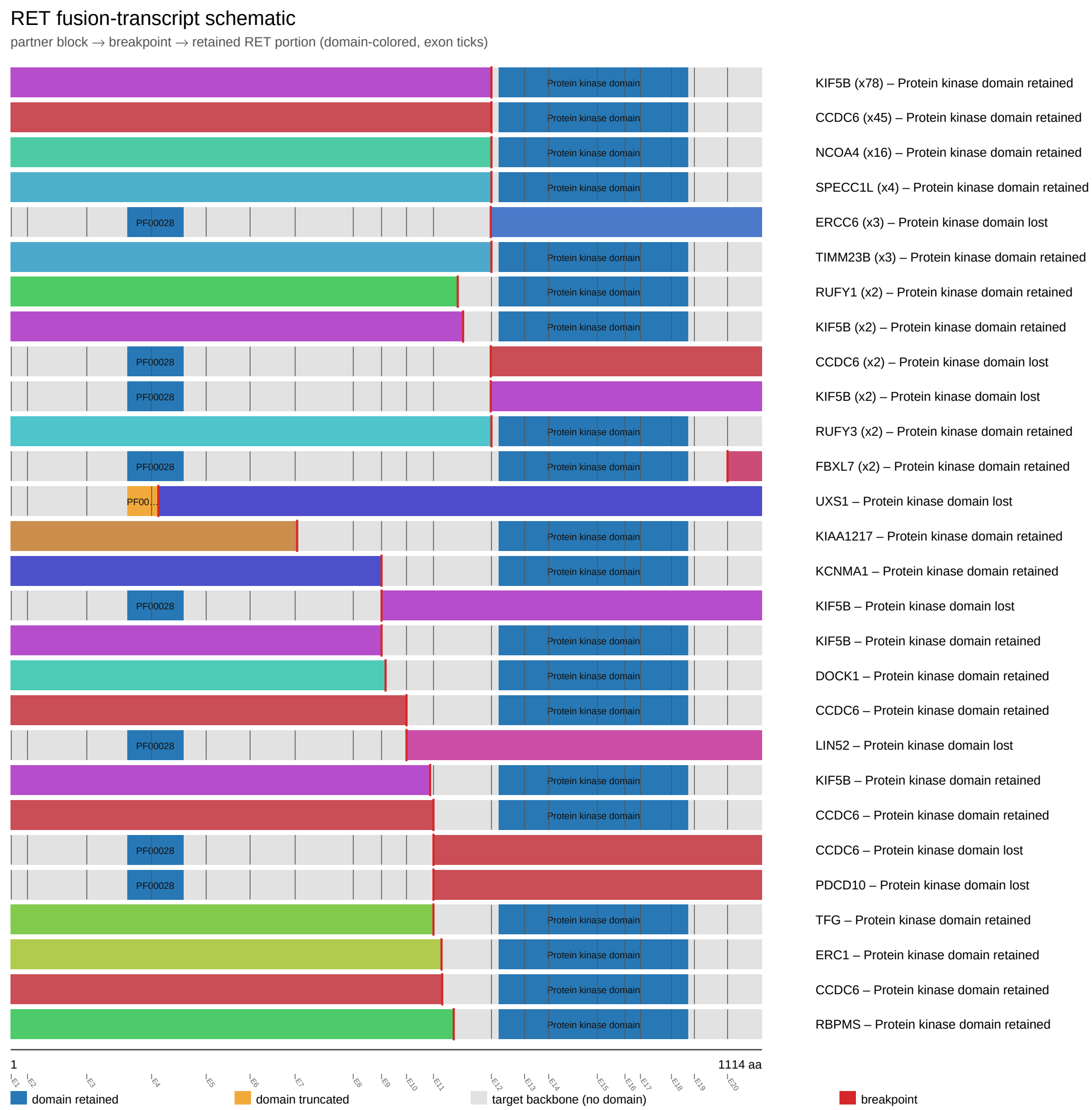


Full per-gene detail: gene_reports/fli1/report.md

RET (Honorable mention, Curated gene config)

RET was analyzed across 194 fusion events, 75.3% in-frame and 92.3% domain-retained. Domain-retention Fisher's exact test p=0.000419666 (raw statistically significant at alpha=0.05). Genome-wide BH-adjusted q=0.096687 (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the RET individual gene report (fusion_schematic.svg).



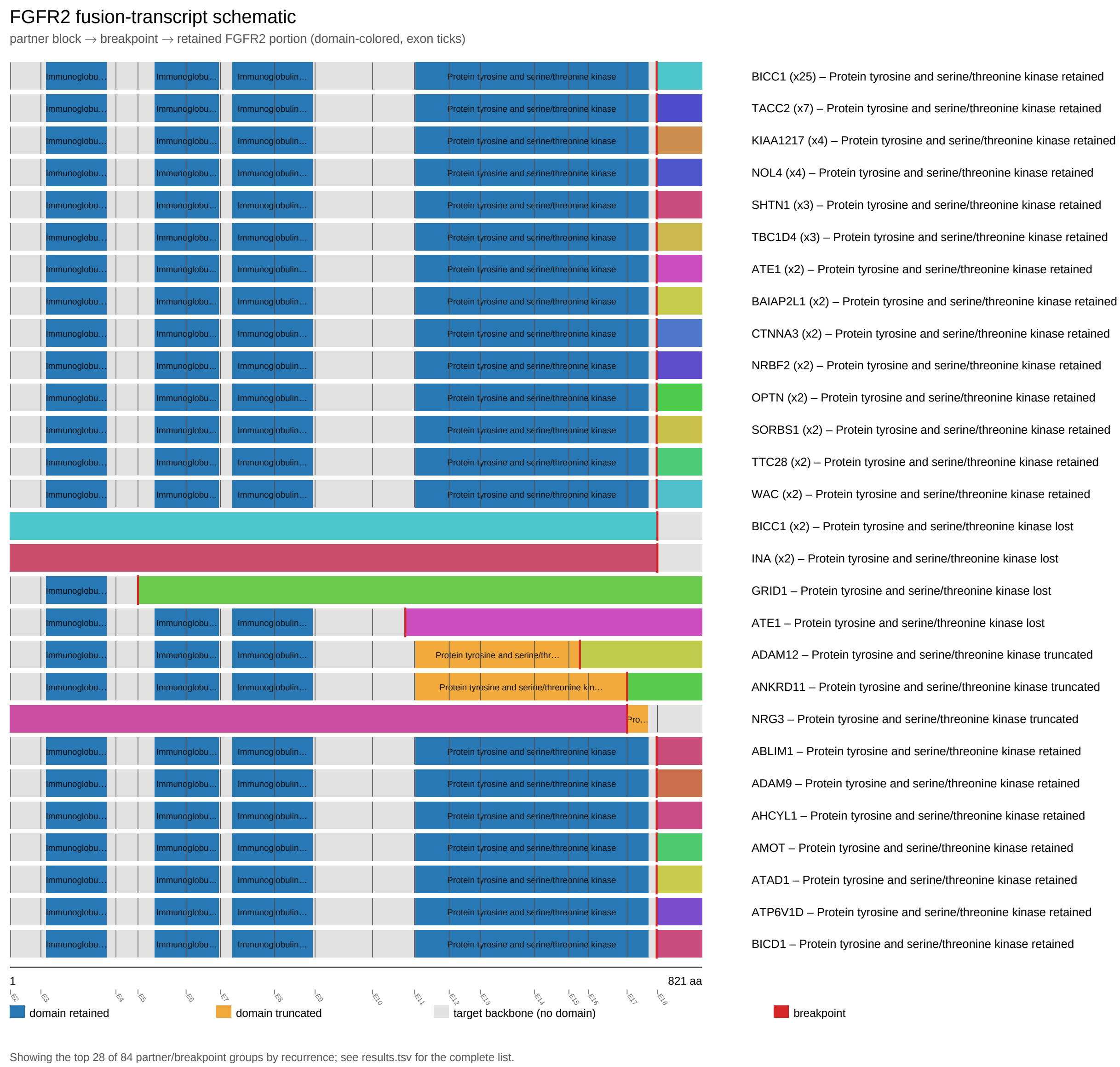
Showing the top 28 of 45 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

Full per-gene detail: gene_reports/ret/report.md

FGFR2 (Honorable mention)

FGFR2 was analyzed across 136 fusion events, 79.4% in-frame and 84.6% domain-retained. Domain-retention Fisher's exact test p=0.000424687 (raw statistically significant at alpha=0.05). Genome-wide BH-adjusted q=0.096687 (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the FGFR2 individual gene report (fusion_schematic.svg).



Full per-gene detail: gene_reports/fgfr2/report.md

ALK (Honorable mention, Curated gene config)

ALK was analyzed across 272 fusion events, 83.1% in-frame and 96.0% domain-retained. Domain-retention Fisher's exact test p=0.00191661 (raw statistically significant at alpha=0.05). Genome-wide BH-adjusted q=0.214679 (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

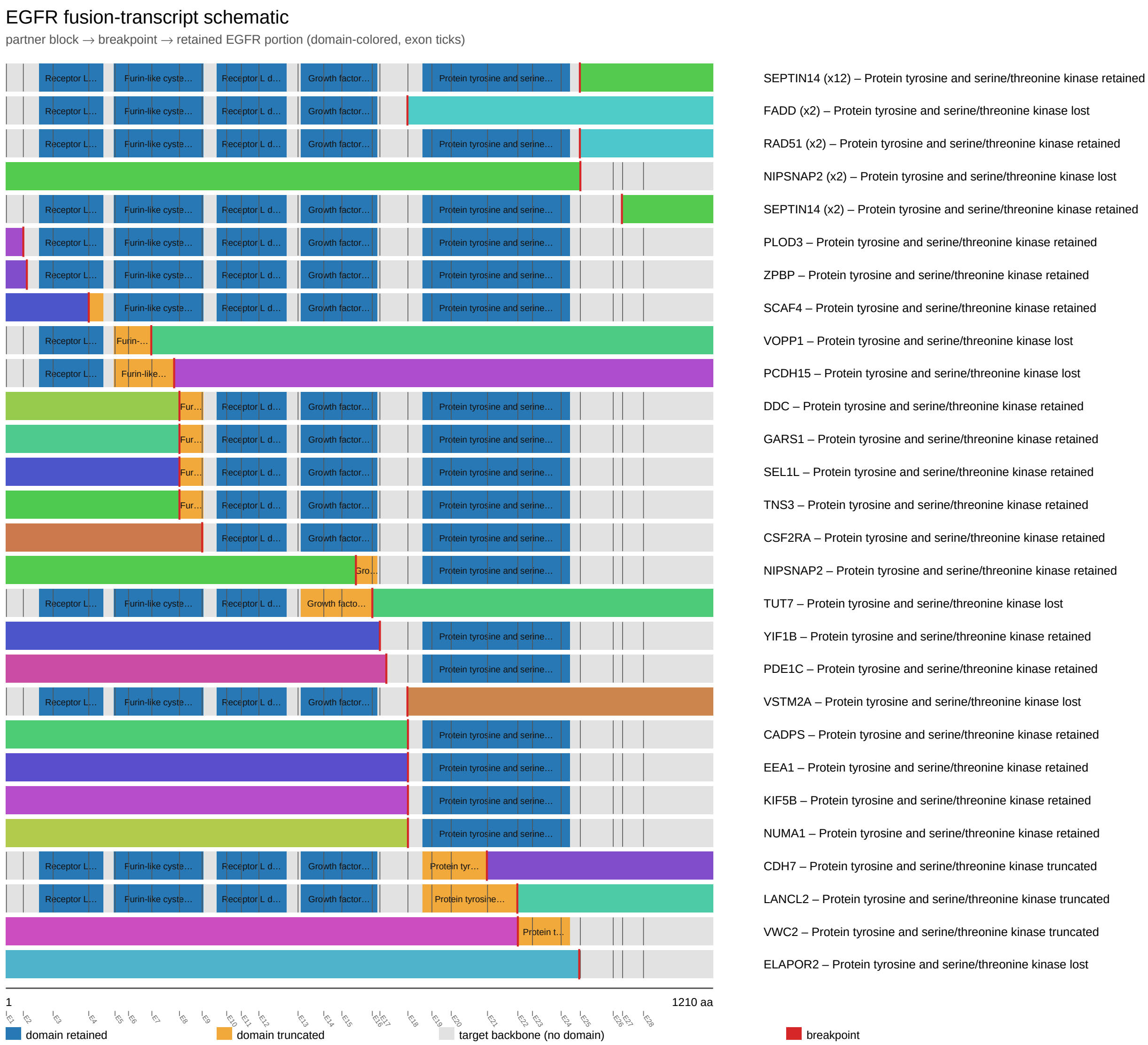
Reused from the ALK individual gene report (fusion_schematic.svg).

Full per-gene detail: gene_reports/alk/report.md

EGFR (Honorable mention)

EGFR was analyzed across 55 fusion events, 50.9% in-frame and 74.5% domain-retained. Domain-retention Fisher's exact test p=0.0114539 (raw statistically significant at alpha=0.05). Genome-wide BH-adjusted q=0.240707 (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the EGFR individual gene report (fusion_schematic.svg).



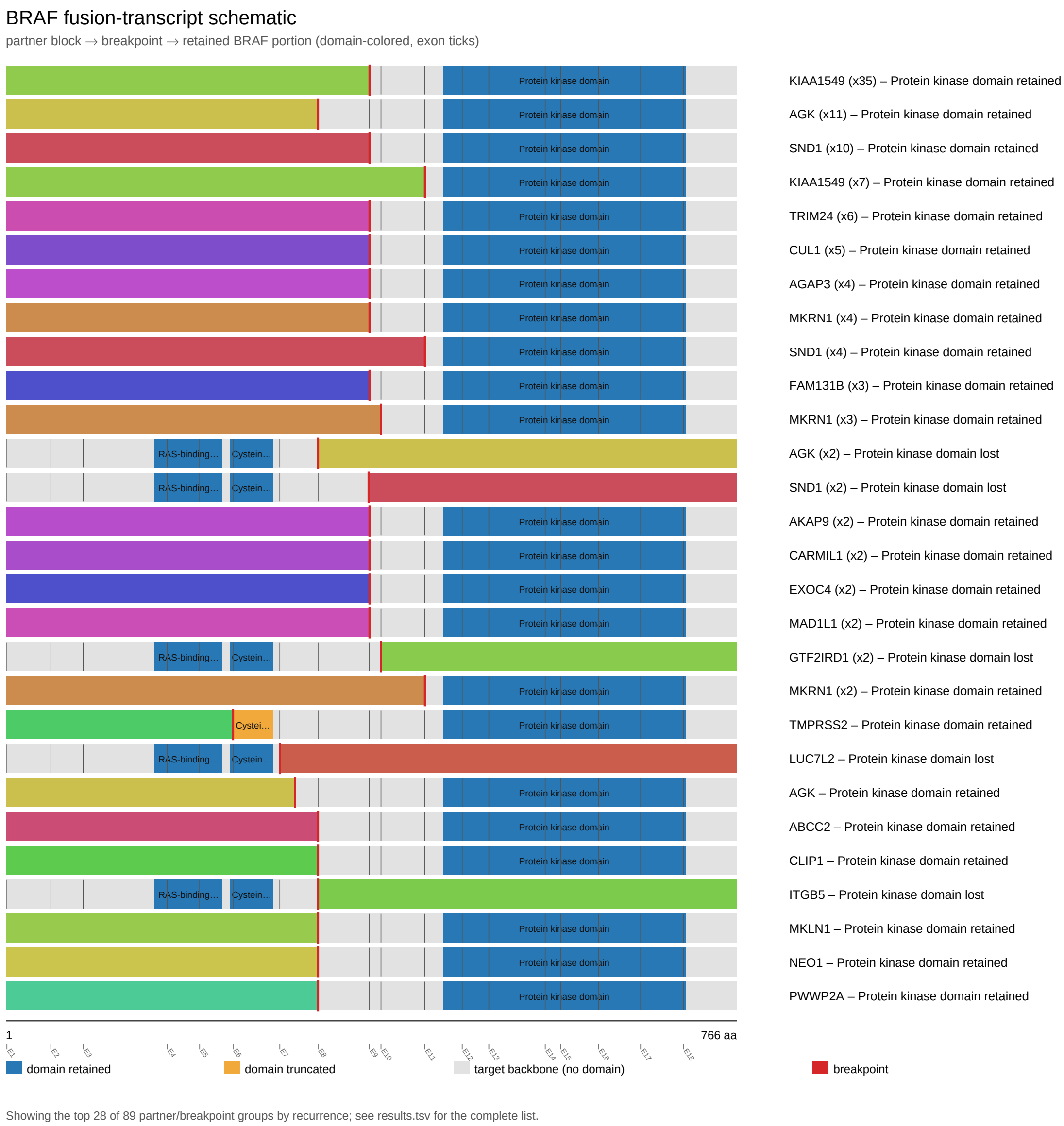
Showing the top 28 of 40 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

Full per-gene detail: gene_reports/egfr/report.md

BRAF (Honorable mention, Curated gene config)

BRAF was analyzed across 179 fusion events, 84.4% in-frame and 91.1% domain-retained. Domain-retention Fisher's exact test p=0.0133676 (raw statistically significant at alpha=0.05). Genome-wide BH-adjusted q=0.214679 (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the BRAF individual gene report (fusion_schematic.svg).



Full per-gene detail: gene_reports/braf/report.md

FGFR3 (Honorable mention)

FGFR3 was analyzed across 152 fusion events, 48.7% in-frame and 93.4% domain-retained. Domain-retention Fisher's exact test p=0.0194824 (raw statistically significant at alpha=0.05). Genome-wide BH-adjusted q=0.214679 (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the FGFR3 individual gene report (fusion_schematic.svg).

FGFR3 fusion-transcript schematic

partner block → breakpoint → retained FGFR3 portion (domain-colored, exon ticks)

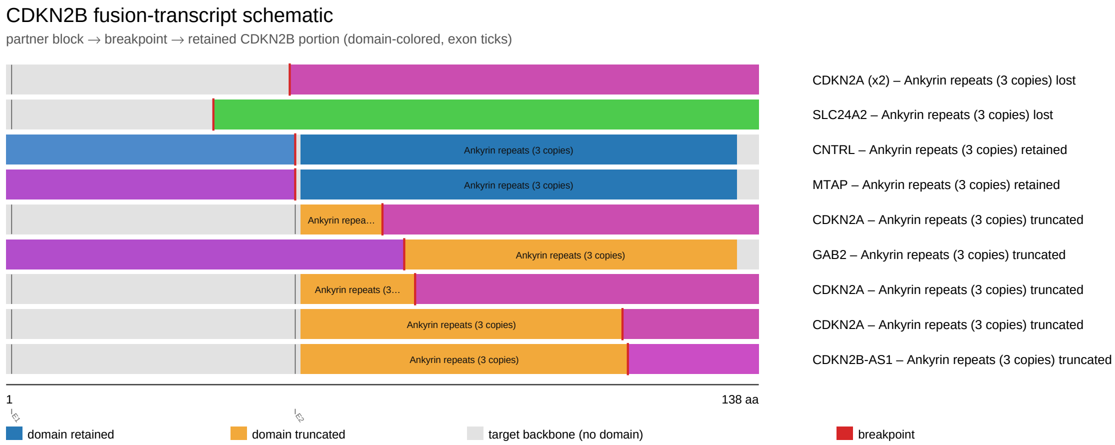
Showing the top 28 of 51 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

Full per-gene detail: gene_reports/fgfr3/report.md

CDKN2B (Honorable mention)

CDKN2B was analyzed across 12 fusion events, 16.7% in-frame and 16.7% domain-retained. Domain-retention Fisher's exact test p=0.0222222 (raw statistically significant at alpha=0.05). Genome-wide BH-adjusted q=0.361376 (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the CDKN2B individual gene report (fusion_schematic.svg).

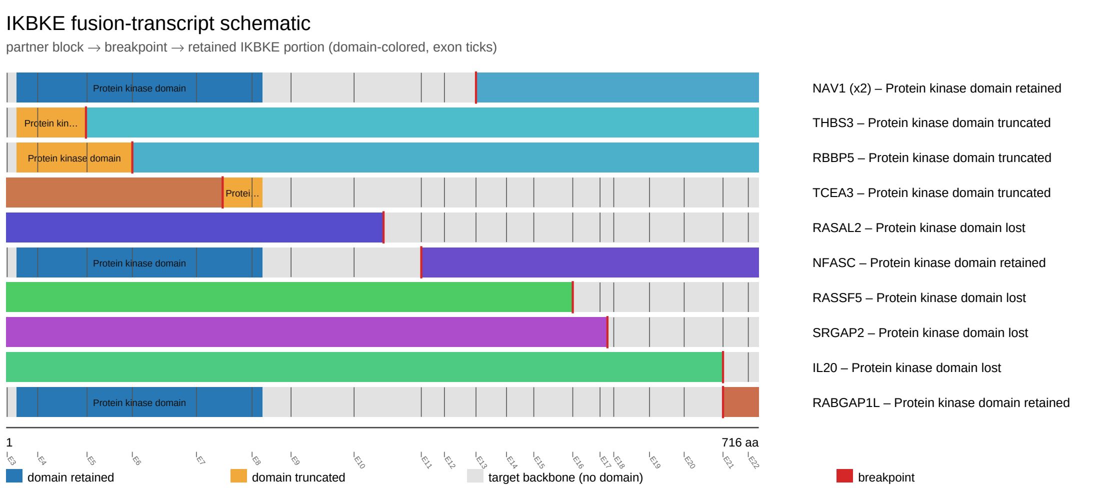


Full per-gene detail: gene_reports/cdkn2b/report.md

IKBKE (Honorable mention)

IKBKE was analyzed across 12 fusion events, 25.0% in-frame and 33.3% domain-retained. Domain-retention Fisher's exact test p=0.0242424 (raw statistically significant at alpha=0.05). Genome-wide BH-adjusted q=0.328828 (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the IKBKE individual gene report (fusion_schematic.svg).

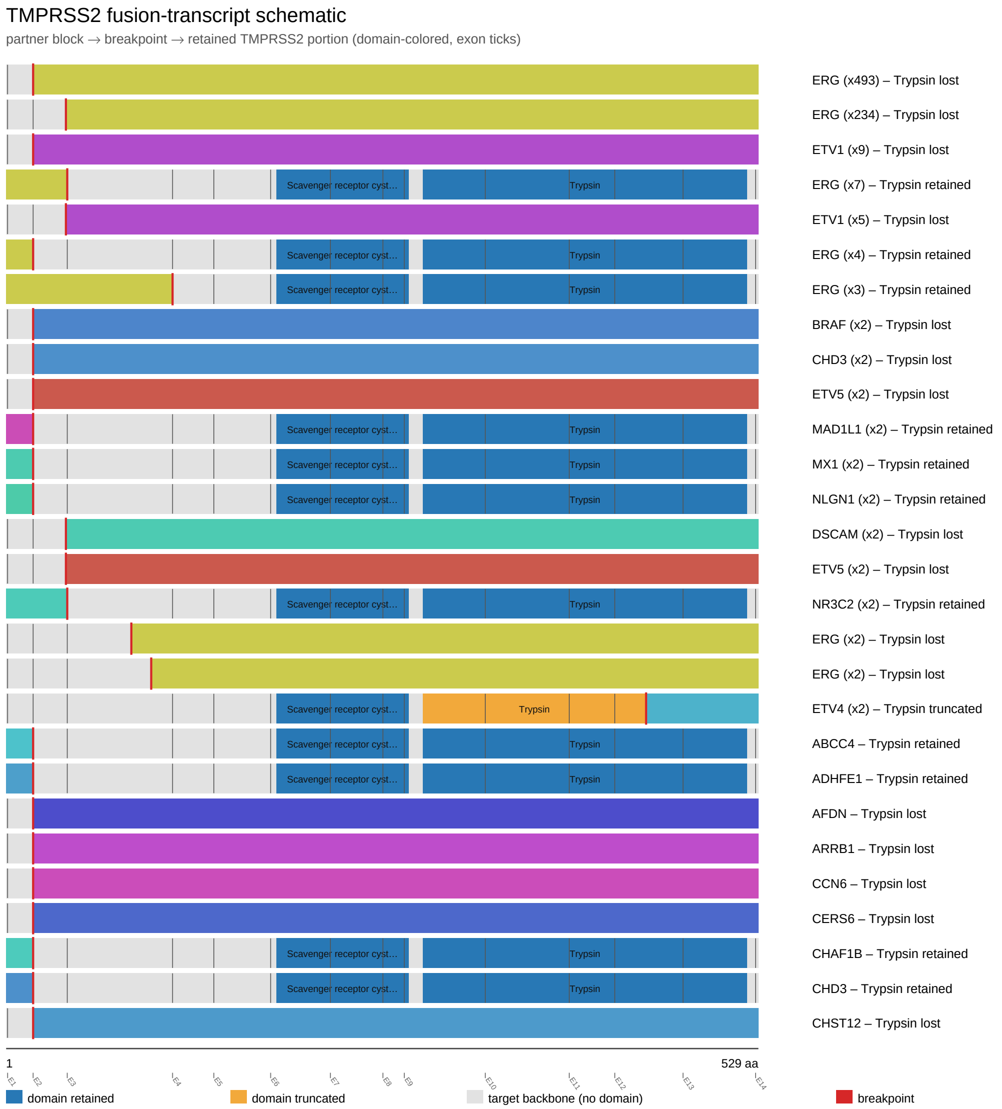


Full per-gene detail: gene_reports/ikbke/report.md

TMPRSS2 (Honorable mention)

TMPRSS2 was analyzed across 867 fusion events, 30.0% in-frame and 6.1% domain-retained. Domain-retention Fisher's exact test p=0.0292045 (raw statistically significant at alpha=0.05). Genome-wide BH-adjusted q=0.214679 (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the TMPRSS2 individual gene report (fusion_schematic.svg).



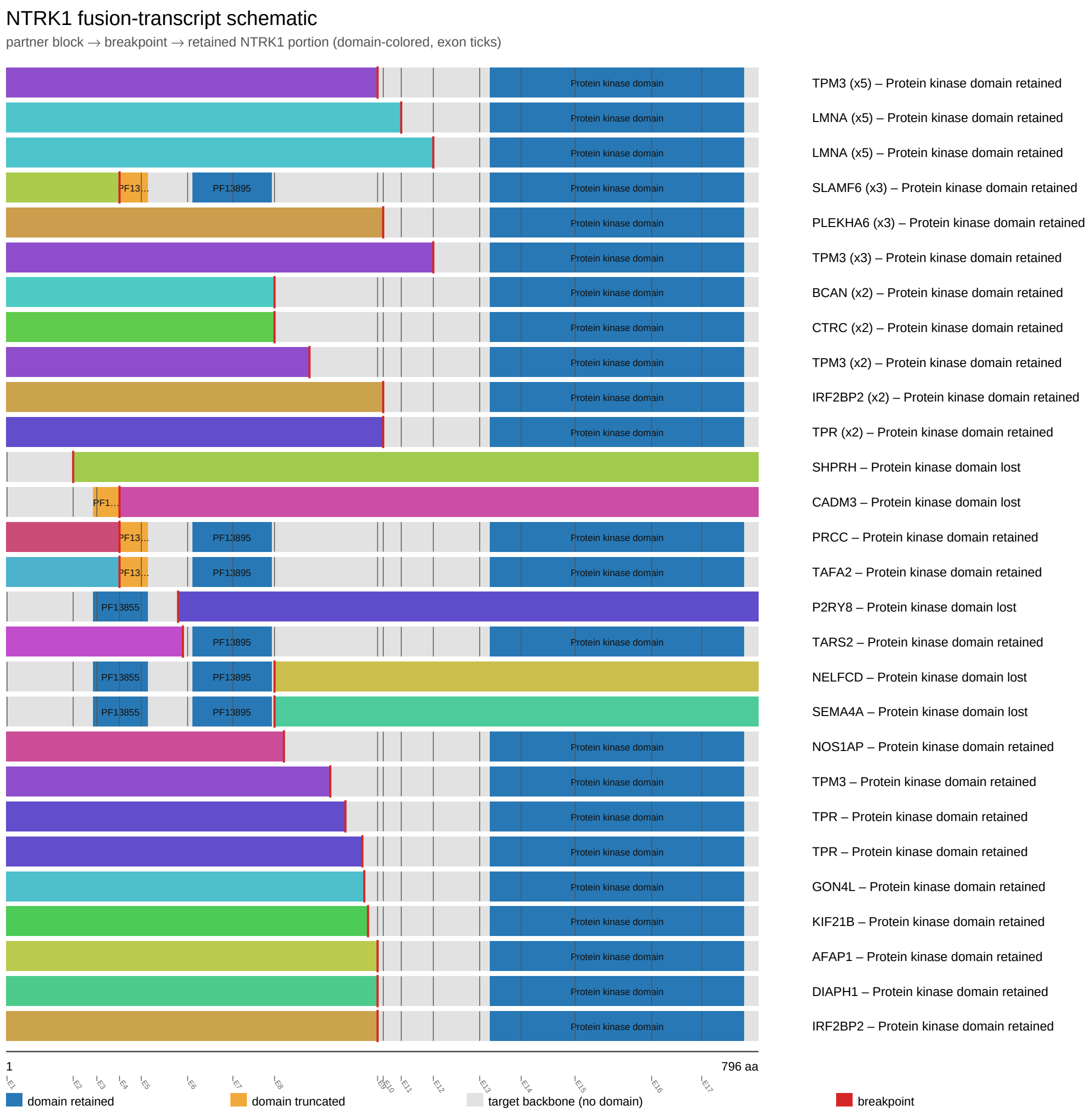
Showing the top 28 of 89 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

Full per-gene detail: gene_reports/tmprss2/report.md

NTRK1 (Honorable mention, Curated gene config)

NTRK1 was analyzed across 78 fusion events, 41.0% in-frame and 82.1% domain-retained. Domain-retention Fisher's exact test p=0.0482628 (raw statistically significant at alpha=0.05). Genome-wide BH-adjusted q=0.214679 (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the NTRK1 individual gene report (fusion_schematic.svg).



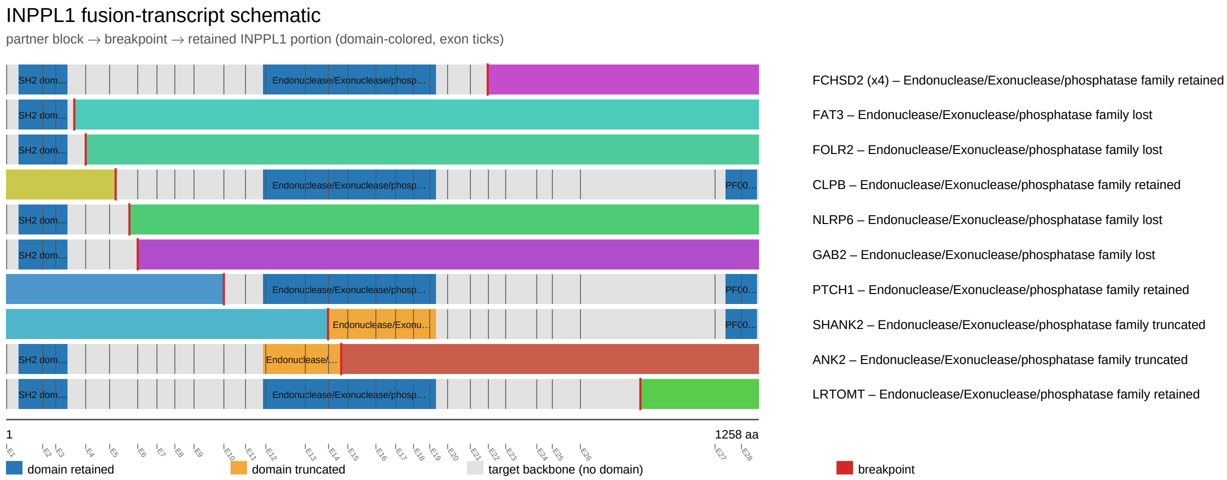
Showing the top 28 of 53 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

Full per-gene detail: gene_reports/ntrk1/report.md

INPPL1 (Honorable mention)

INPPL1 was analyzed across 14 fusion events, 28.6% in-frame and 50.0% domain-retained. Domain-retention Fisher's exact test p=0.048951 (raw statistically significant at alpha=0.05). Genome-wide BH-adjusted q=0.468663 (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the INPPL1 individual gene report (fusion_schematic.svg).

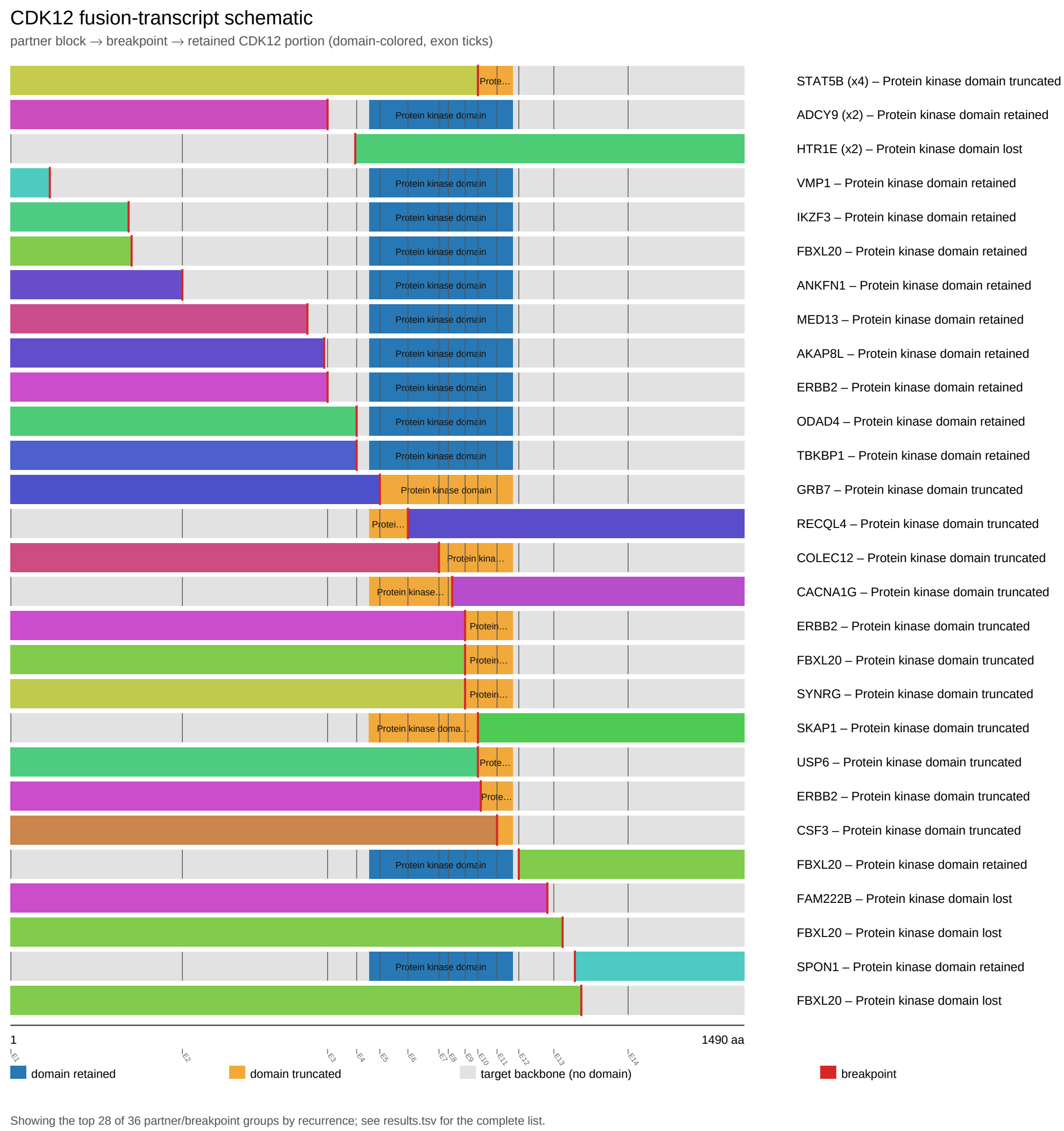


Full per-gene detail: gene_reports/inppl1/report.md

CDK12 (Honorable mention)

CDK12 was analyzed across 43 fusion events, 16.3% in-frame and 37.2% domain-retained. Domain-retention Fisher's exact test p=0.0677006 (raw not statistically significant at alpha=0.05). Genome-wide BH-adjusted q=0.214679 (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the CDK12 individual gene report (fusion_schematic.svg).

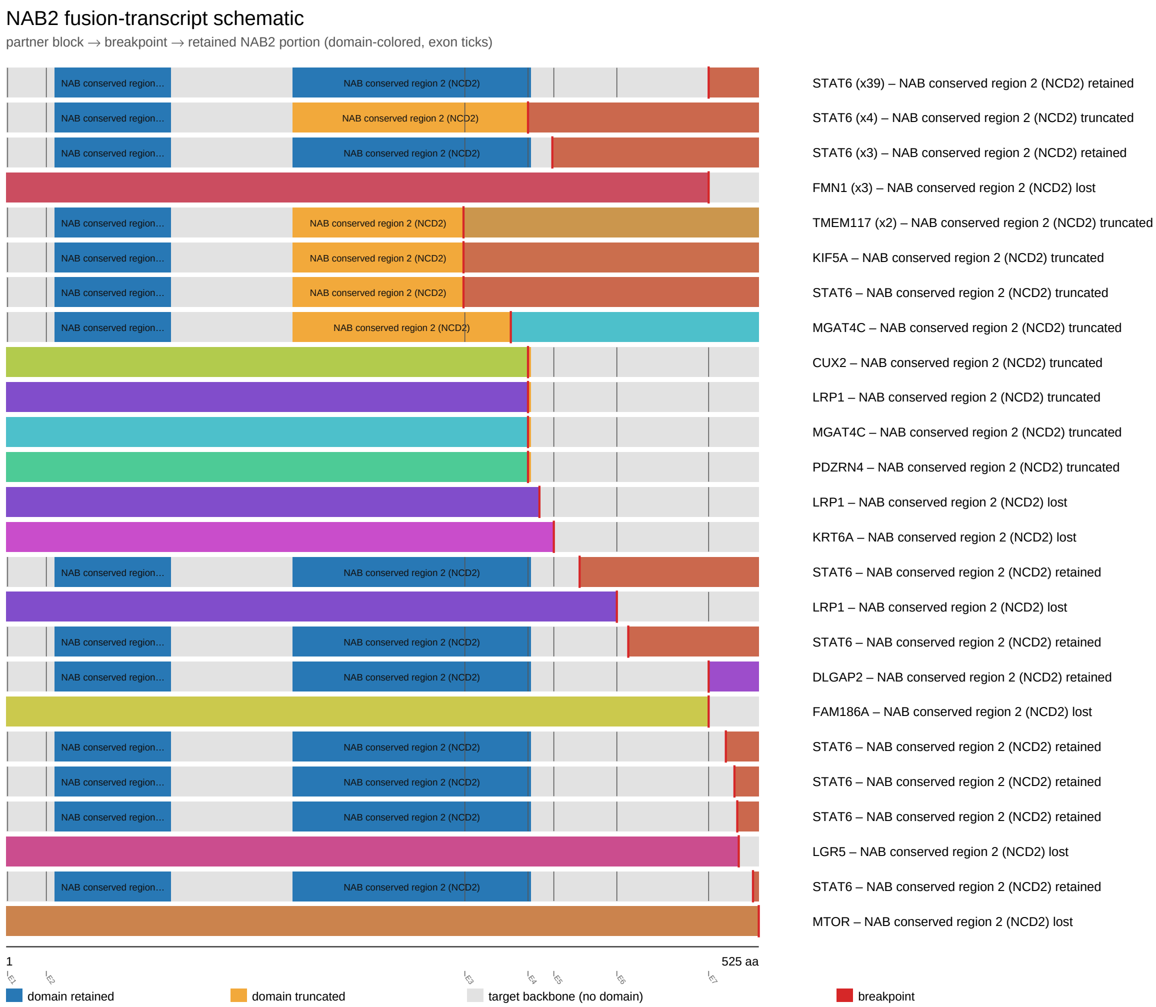


Full per-gene detail: gene_reports/cdk12/report.md

NAB2 (Honorable mention)

NAB2 was analyzed across 72 fusion events, 34.7% in-frame and 68.1% domain-retained. Domain-retention Fisher's exact test p=0.11258 (raw not statistically significant at alpha=0.05). Genome-wide BH-adjusted q=0.214679 (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the NAB2 individual gene report (fusion_schematic.svg).

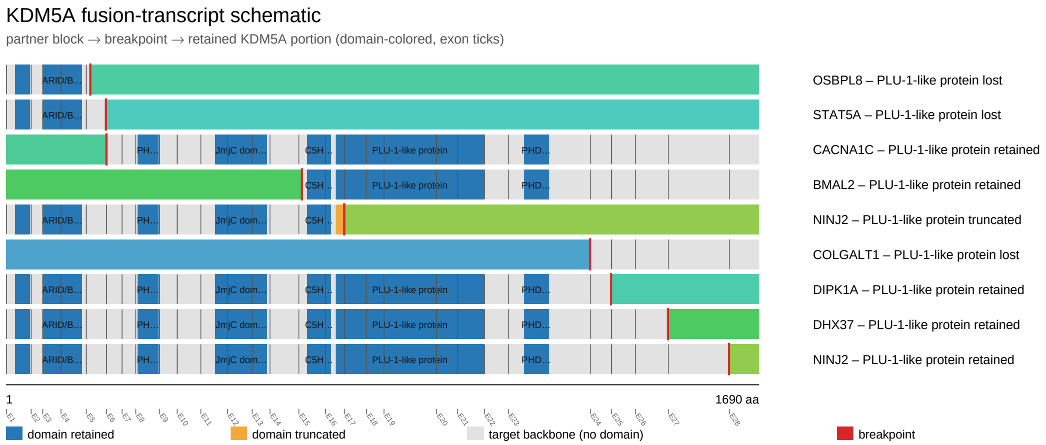


Full per-gene detail: gene_reports/nab2/report.md

KDM5A (Honorable mention)

KDM5A was analyzed across 10 fusion events, 30.0% in-frame and 50.0% domain-retained. Domain-retention Fisher's exact test p=0.119048 (raw not statistically significant at alpha=0.05). Genome-wide BH-adjusted q=0.878723 (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the KDM5A individual gene report (fusion_schematic.svg).

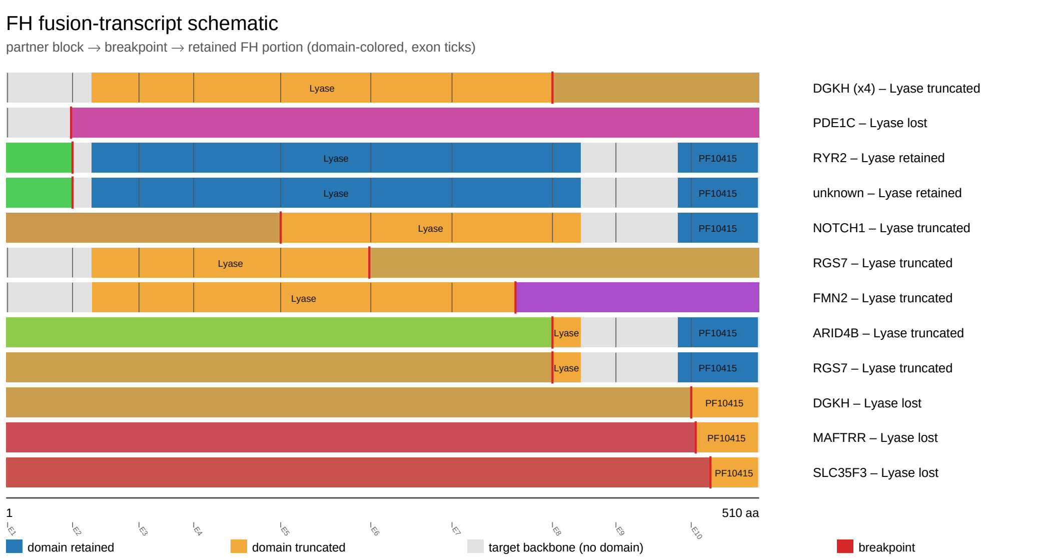


Full per-gene detail: gene_reports/kdm5a/report.md

FH (Honorable mention)

FH was analyzed across 16 fusion events, 6.2% in-frame and 12.5% domain-retained. Domain-retention Fisher's exact test p=0.133333 (raw not statistically significant at alpha=0.05). Genome-wide BH-adjusted q=0.671921 (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the FH individual gene report (fusion_schematic.svg).



Full per-gene detail: gene_reports/fh/report.md

Discussion

• Cross-gene Benjamini-Hochberg FDR correction was applied jointly across the 359 genes that produced at least one computable p-value in this scan. This reduces false-positive findings relative to testing each gene in isolation, but is a conservative correction: a real per-gene effect can fail to reach the q < 0.05 threshold once corrected across that full p-value-bearing gene set.

• 4 gene(s) used hand-curated gene configurations, while 520 gene(s) were auto-configured from Genome Nexus canonical-transcript/Pfam-domain data using a kinase/catalytic-keyword heuristic to select the tracked domain; auto-configured domains have not been manually verified the way hand-curated ones have.

• Data were retrieved live from the public msk_impact_50k_2026 cBioPortal study as of 2026-09-05T01:41:00.493319+00:00. Some cBioPortal cohorts (e.g. actively accruing clinical-sequencing panels such as MSK-IMPACT) are updated periodically, so exact counts could shift if this scan is re-run later against such a cohort.

• All findings in this manuscript are computational, hypothesis-generating candidate evidence from bioinformatic analysis of public cohort data, not validated clinical calls; a gene's presence here is not a therapeutic or diagnostic recommendation.

Appendix: per-gene report index

Gene	Tier	Config source	N events	Best q	Full report
NTRK3	FDR-significant	auto	58	8.165e-09	gene_reports/ntrk3/report.md
ROS1	FDR-significant	auto	122	0.0002213	gene_reports/ros1/report.md
ETV6	FDR-significant	auto	90	0.002333	gene_reports/etv6/report.md
FLI1	FDR-significant	auto	118	0.008206	gene_reports/fli1/report.md
RET	Honorable mention; Curated gene config	curated	194	0.09669	gene_reports/ret/report.md
FGFR2	Honorable mention	auto	136	0.09669	gene_reports/fgfr2/report.md
ALK	Honorable mention; Curated gene config	curated	272	0.2147	gene_reports/alk/report.md
EGFR	Honorable mention	auto	55	0.2407	gene_reports/egfr/report.md
BRAF	Honorable mention; Curated gene config	curated	179	0.2147	gene_reports/braf/report.md
FGFR3	Honorable mention	auto	152	0.2147	gene_reports/fgfr3/report.md
CDKN2B	Honorable mention	auto	12	0.3614	gene_reports/cdkn2b/report.md
IKBKE	Honorable mention	auto	12	0.3288	gene_reports/ikbke/report.md
TMPRSS2	Honorable mention	auto	867	0.2147	gene_reports/tmprss2/report.md
NTRK1	Honorable mention; Curated gene config	curated	78	0.2147	gene_reports/ntrk1/report.md
INPPL1	Honorable mention	auto	14	0.4687	gene_reports/inppl1/report.md
CDK12	Honorable mention	auto	43	0.2147	gene_reports/cdk12/report.md
NAB2	Honorable mention	auto	72	0.2147	gene_reports/nab2/report.md
KDM5A	Honorable mention	auto	10	0.8787	gene_reports/kdm5a/report.md
FH	Honorable mention	auto	16	0.6719	gene_reports/fh/report.md